

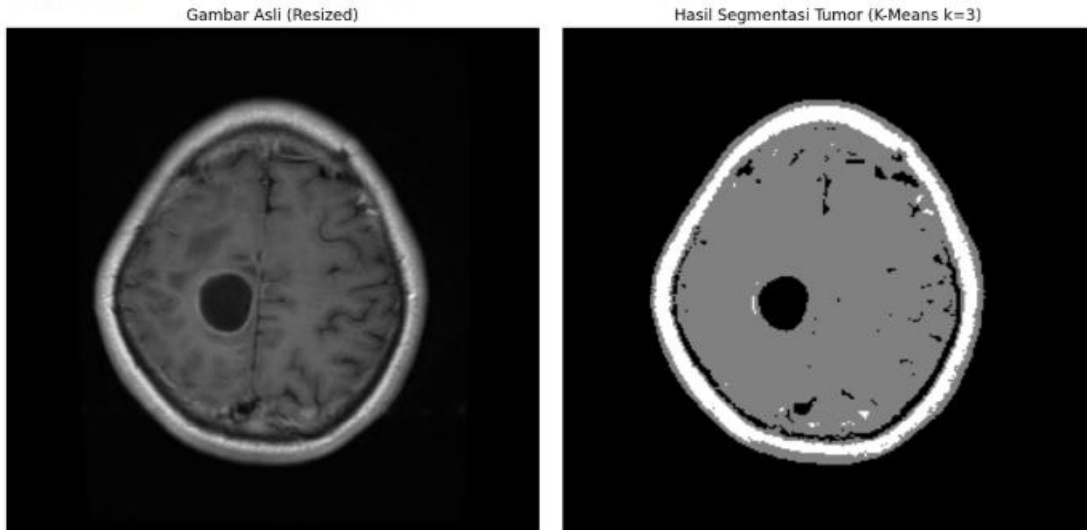
## A. Hasil Running

### 1. Dataset yang Digunakan

```
Using Colab cache for faster access to the 'brian-tumor-dataset' dataset.  
Path ke file dataset: /kaggle/input/brian-tumor-dataset  
Total gambar ditemukan: 4600
```

### 2. Hasil Segmentasi Tumor

Converged in 16 iteration



### 3. Split Data Training dan Testing

```
Total images: 4434  
Training images: 3990  
Testing images: 444  
Training Tumor ratio: 0.53  
Testing Tumor ratio: 0.53
```

Mengekstrak centroid untuk data training secara PARALEL...

```
Train Centroids: 100% ██████████ 3990/3990 [03:29<00:00, 20.37it/s]
```

Berhasil mengekstrak centroid untuk 3990 gambar training.

Mengekstrak centroid untuk data testing secara PARALEL...

```
Test Centroids: 100% ██████████ 444/444 [00:22<00:00, 23.61it/s]
```

Berhasil mengekstrak centroid untuk 444 gambar testing.

### 4. Distribusi Data pada CPU saat Proses Paralelisasi

```
Train Centroids: 100% ██████████ 3990/3990 [03:37<00:00, 20.18it/s]
```

```
=== DISTRIBUSI DATA PADA INTI CPU (TRAINING) ===  
Total Inti CPU (Worker) Aktif: 2  
Inti CPU ke-1 (PID: 32125) -> Memproses 2025 gambar  
Inti CPU ke-2 (PID: 32126) -> Memproses 1965 gambar
```

```
--- Distribusi data pada CPU (KLASIFIKASI KNN) ---  
Inti CPU ke-1 (PID: 42532) -> Mengklasifikasikan 222 gambar  
Inti CPU ke-2 (PID: 42531) -> Mengklasifikasikan 222 gambar
```

## 5. Hasil Pengujian pada Modified KNN Classification

Melakukan klasifikasi pada data test...

Parallel Modified KNN Classification: 100%  444/444 [01:18<00:00, 6.41it/s]

Evaluation Metrics

Accuracy: 0.9414

Precision: 0.9646

Recall: 0.9237

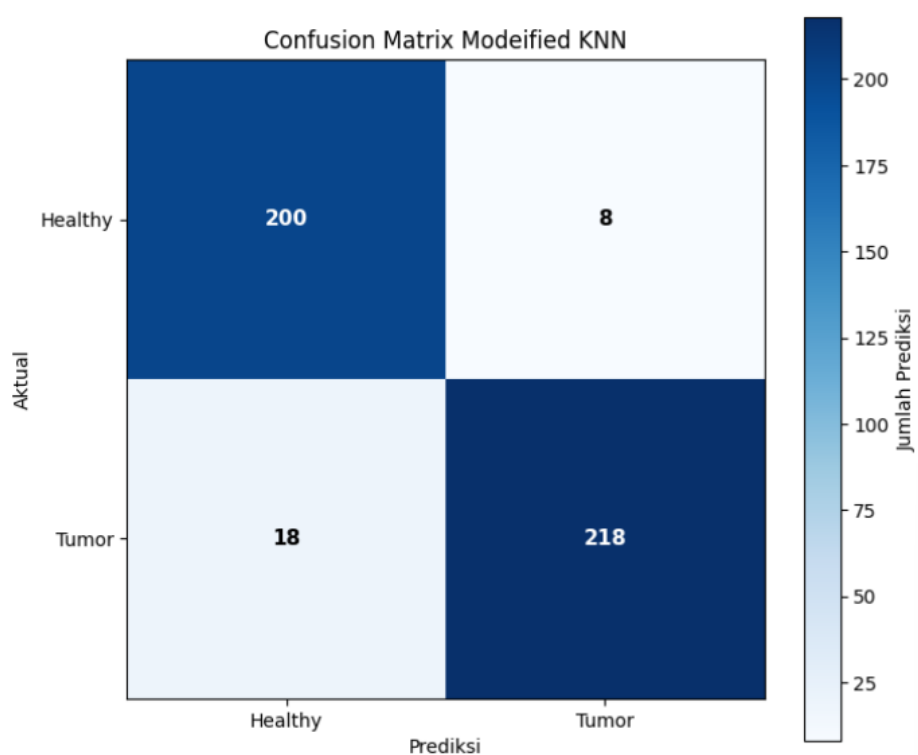
F1 Score: 0.9437

Confusion Matrix:

```
[[200  8]
```

```
[ 18 218]]
```

## 6. Visualisasi Confusion Matrix pada Modified KNN Classification



## 7. Hasil Pengujian pada KNN Library Classification

Training KNN dengan Library...

Melakukan klasifikasi secara PARALEL dengan KNN Library...

Parallel KNN Library Classification: 100%  444/444 [00:04<00:00, 130.62it/s]

Evaluation Metrics

Accuracy : 0.7140

Precision: 0.7444

Recall : 0.7034

F1 Score : 0.7233

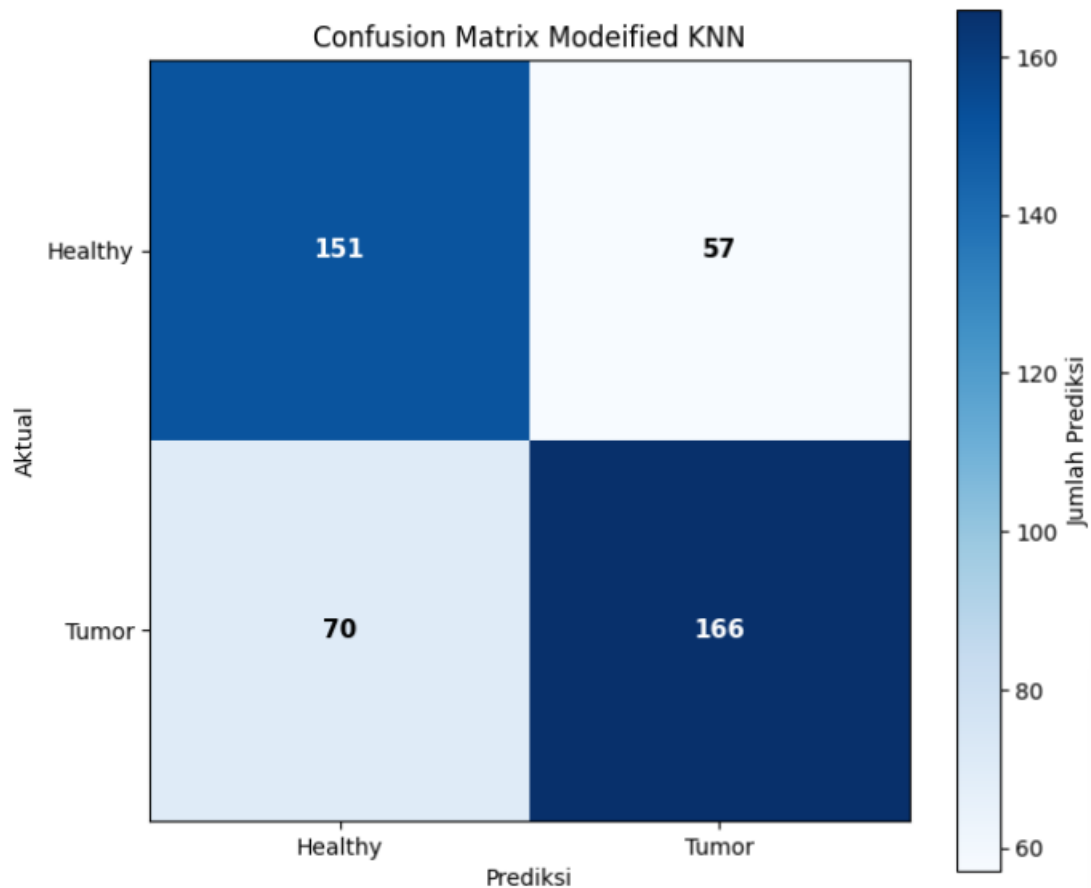
Confusion Matrix:

```
[[151  57]
```

```
[ 70 166]]
```

Execution Time: 5.4173 seconds

## 8. Visualisasi Confusion Matrix KNN Library Classification



## 9. Hasil Pengujian Modified KNN Classification pada Dataset Lain

```
base_new_path = test_path + "/Dataset/Brain Tumor MRI images"
print(base_new_path)
```

---

/root/.cache/kagglehub/datasets/murtozalikhon/brain-tumor-multimodal-image-ct-and-mri/versions/1/Dataset/Brain Tumor MRI images

Found 3000 tumor images and 2000 healthy images.  
Total images: 5000

Mengekstrak centroid untuk data baru secara PARALEL...

Test Centroids: 100%  5000/5000 [04:09<00:00, 13.92it/s]

Berhasil mengekstrak centroid untuk 5000 gambar testing.

Processing new dataset and classifying in PARALEL...

Parallel Modified KNN Classification New Data Test: 92%  4584/5000 [13:56<01:09, 5.95it/s]

Evaluation Metrics

Accuracy: 0.8640

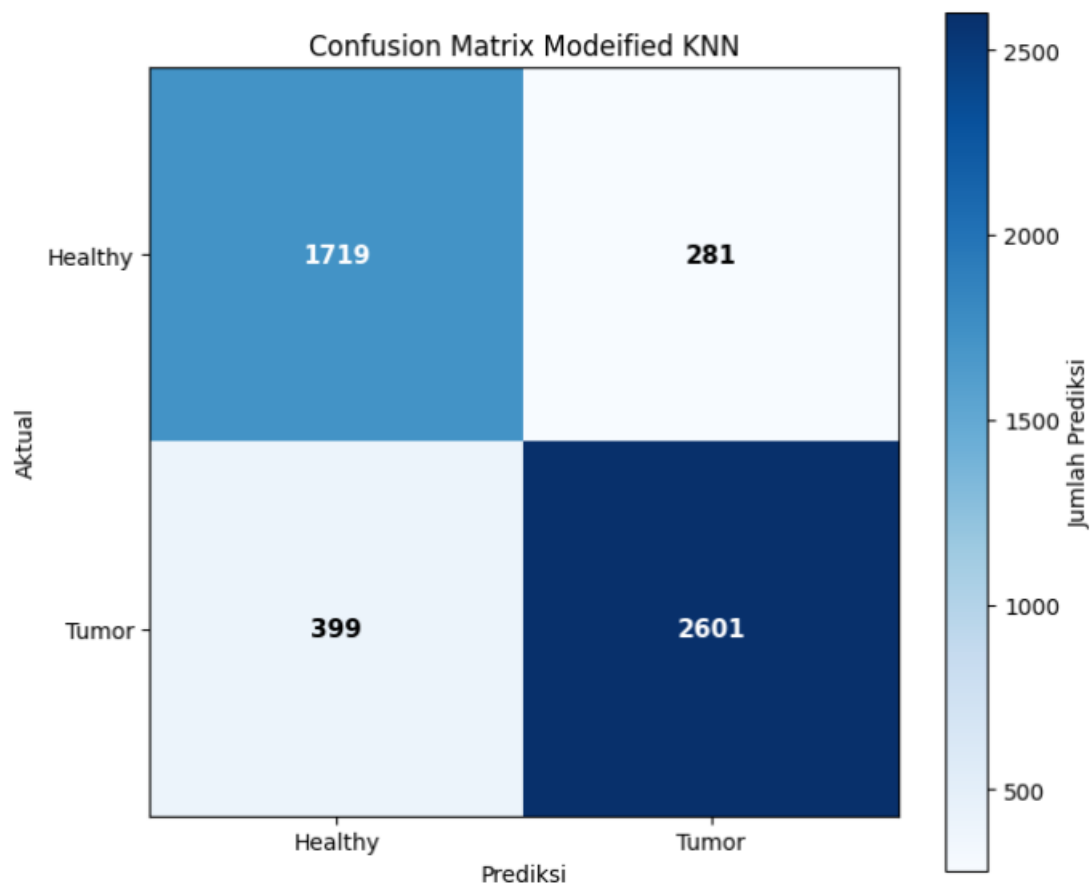
Confusion Matrix:

```
[[1719 281]
 [ 399 2601]]
```

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Healthy      | 0.81      | 0.86   | 0.83     | 2000    |
| Cancer       | 0.90      | 0.87   | 0.88     | 3000    |
| accuracy     |           |        | 0.86     | 5000    |
| macro avg    | 0.86      | 0.86   | 0.86     | 5000    |
| weighted avg | 0.87      | 0.86   | 0.86     | 5000    |

# 10. Visualisasi Confusion Matrix Modified KNN Classification pada Dataset Lain



## 11. Visualisasi Persamaan $w = \frac{1}{d+\epsilon}$ , dimana $\epsilon = 10^{-6}$

